FINANCIAL INNOVATION

IMPLICATIONS FOR PAYMENTS AND POLICY

FEDERAL RESERVE BANK OF KANSAS CITY
JACKSON HOLE, WYOMING
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FEDERAL RESERVE BANK OF KANSAS CITY

Financial Innovation and the International Monetary System

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- Logic vs. reality
- Will financial innovation reshape the IMS?
- How might innovation reshape the IMS?
- A liquidity-based model of debt determination
- Policy implications (for rest of the world)

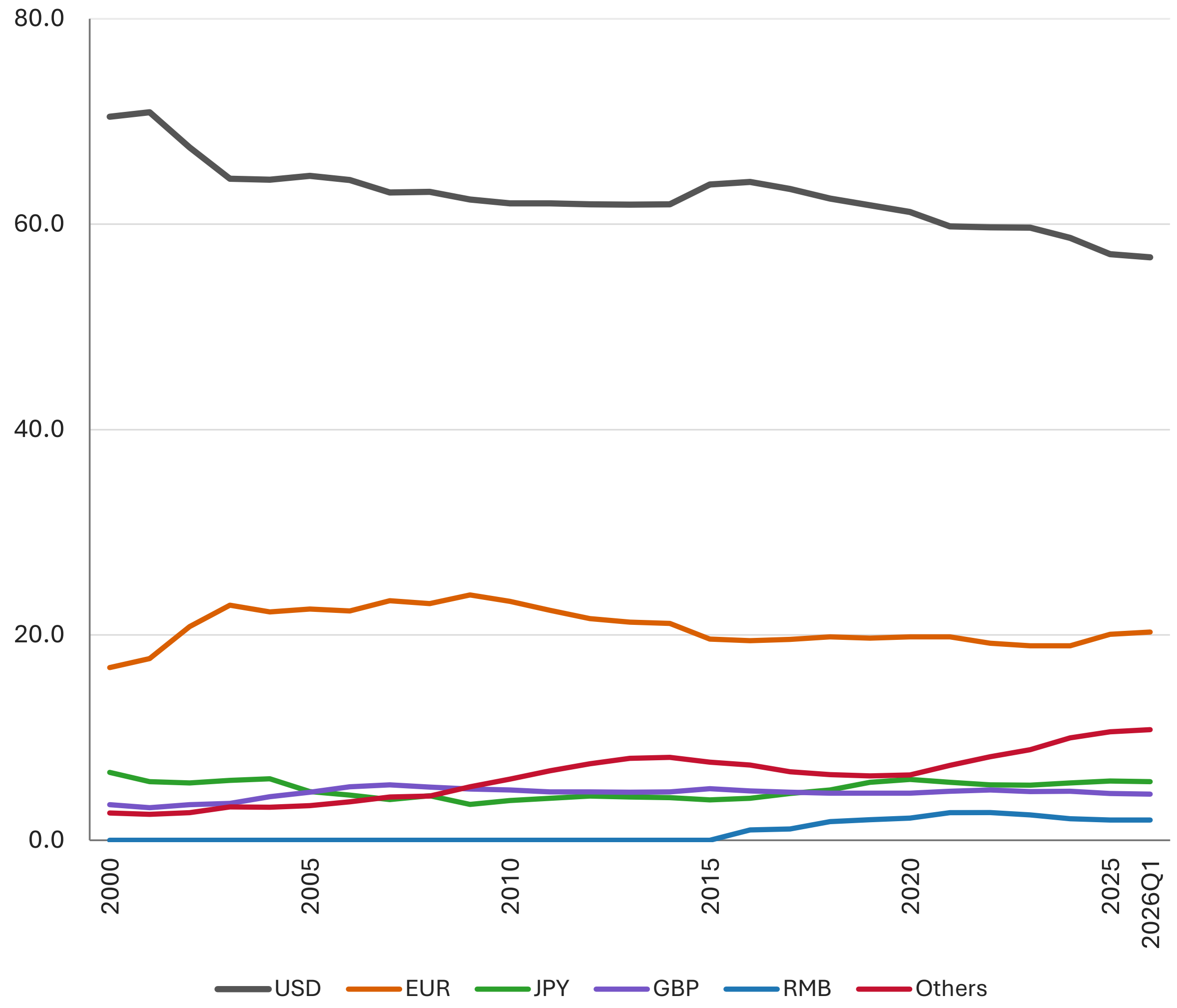
Logic vs. Reality

- Dollar dominance declining; safe asset status of $/US Treasuries in question
  - Logical: Macro policies, erosion of institutions, weaponization thru sanctions
  - Evidence: FX reserve shares, convenience yields
- Dollar (and U.S. assets) firming up while other currencies stumble/stall
  - Payments, FX turnover, debt issuance
  - Share of global external wealth allocated to U.S. assets
  - EMDEs & SOEs: external asset accumulation by private sector
- Multipolarity vs fragmentation
  - New entrants: RMB, smaller reserve currencies
  - Traditional reserve currencies (EUR, JPY, GBP) fading in relative terms
  - Intensified competition in second tier

The Dollar Loses Ground

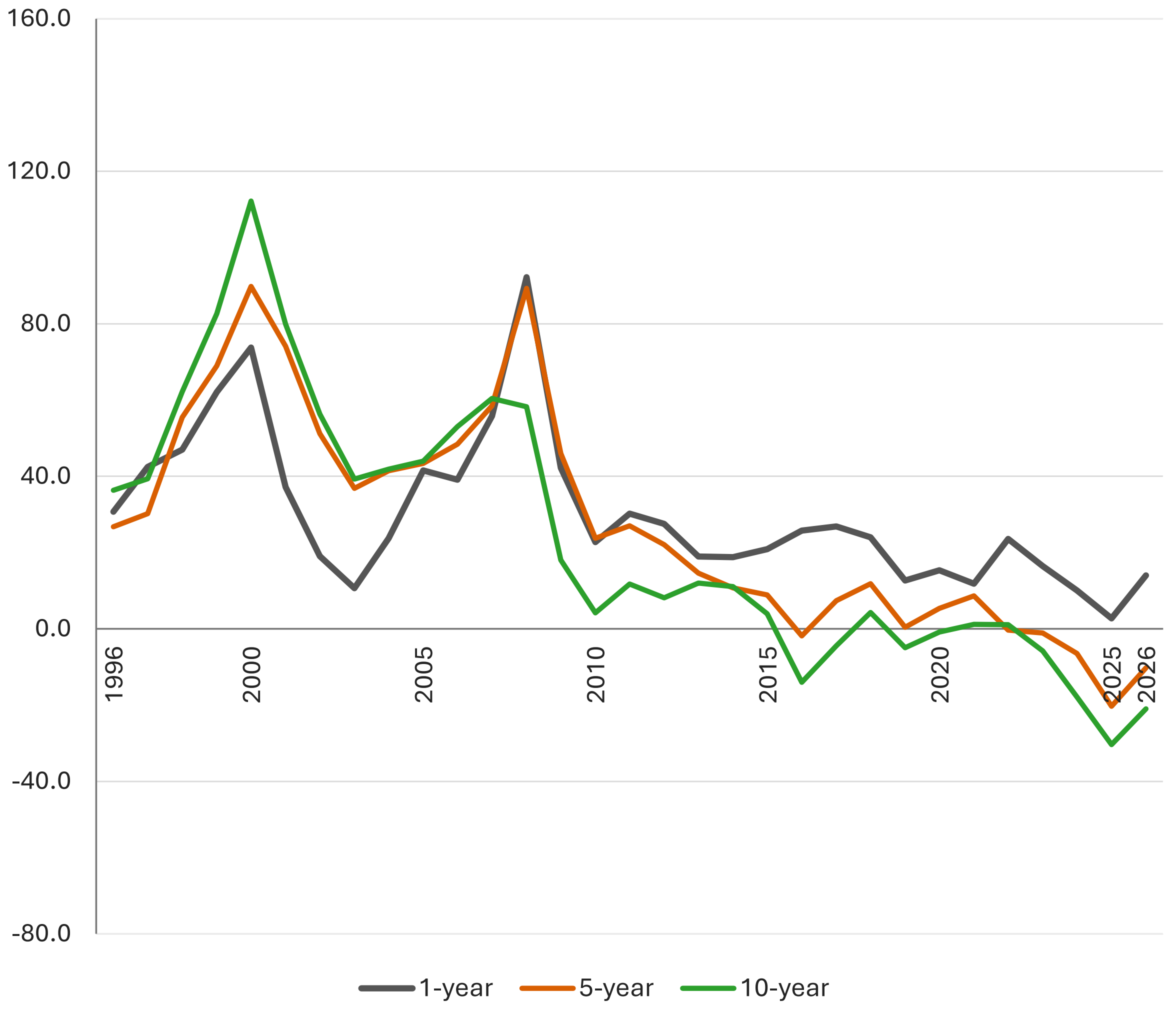
FX Reserve Shares

(in percent)



Measures of Convenience Yield

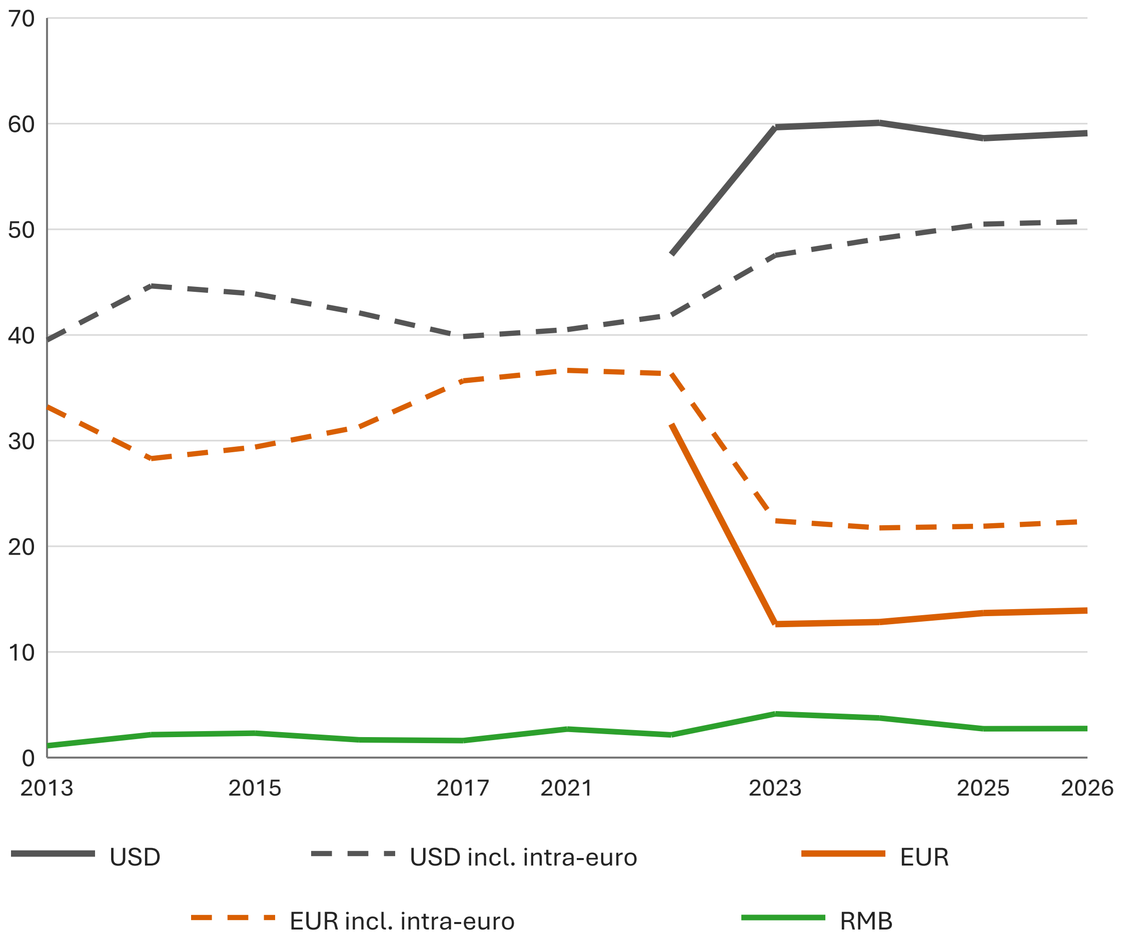
(basis points)



Dollar Dominance Persists

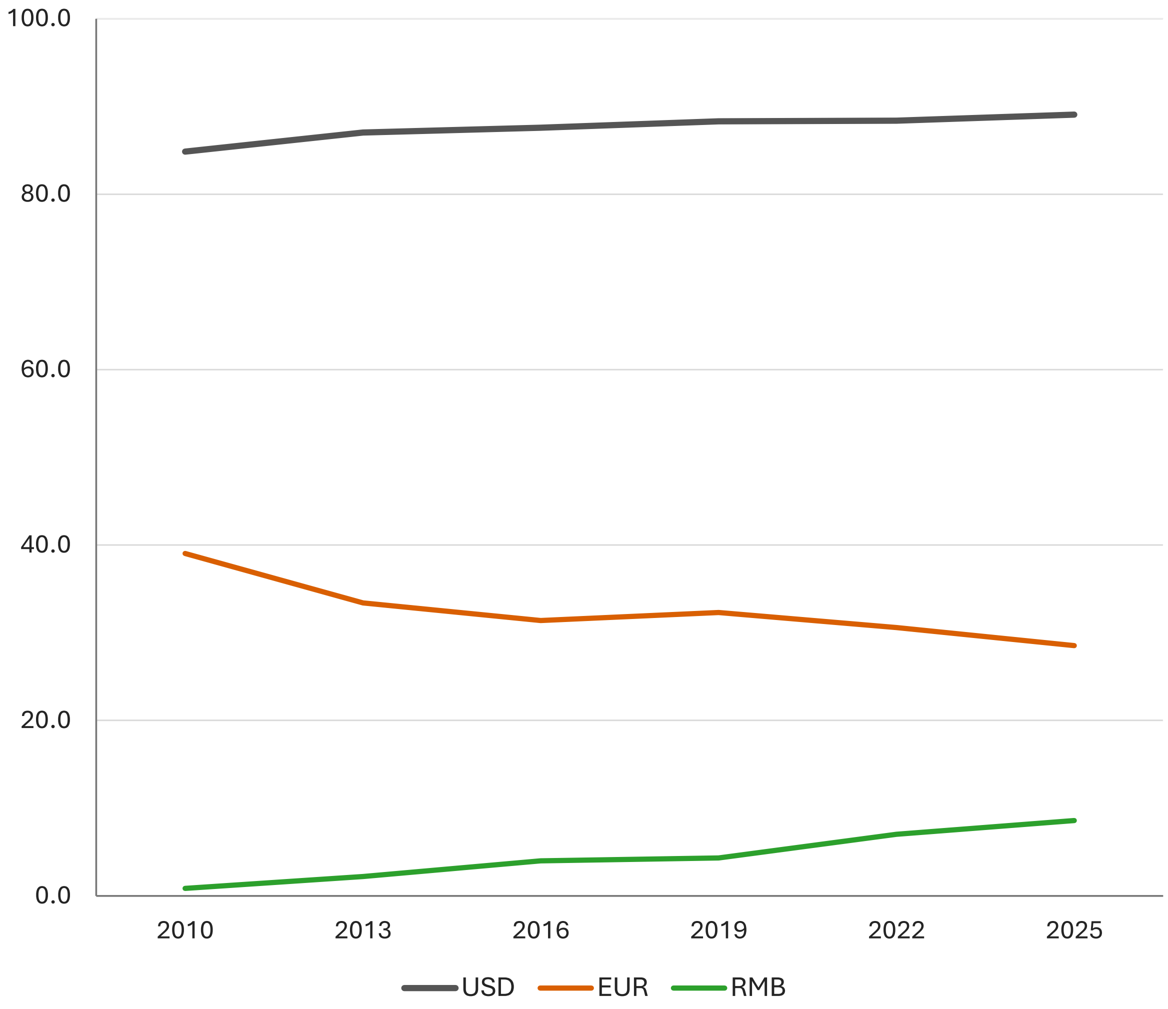
International Payments (SWIFT)

(currency shares, in percent)



FX Turnover

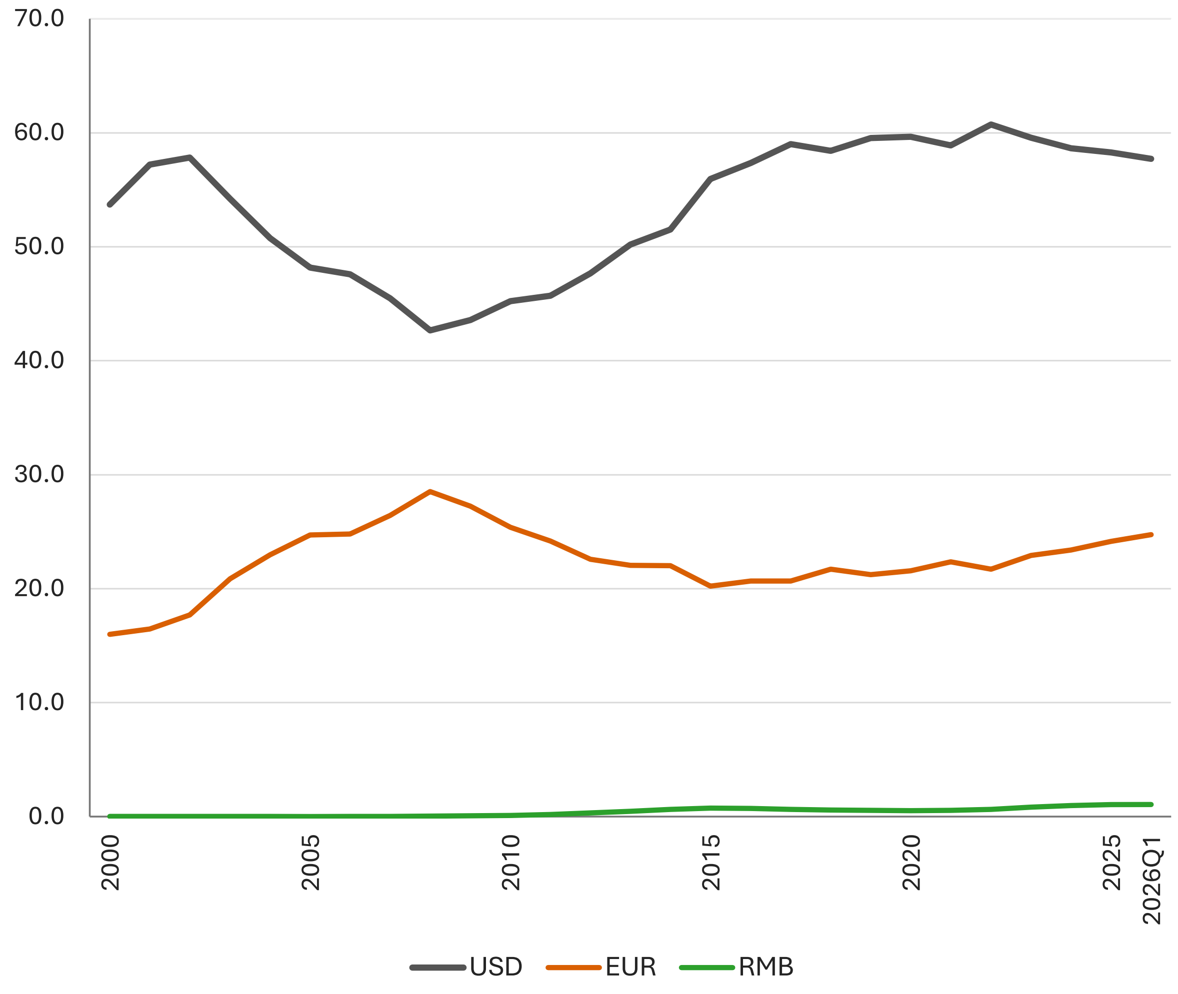
(currency shares, in percent)



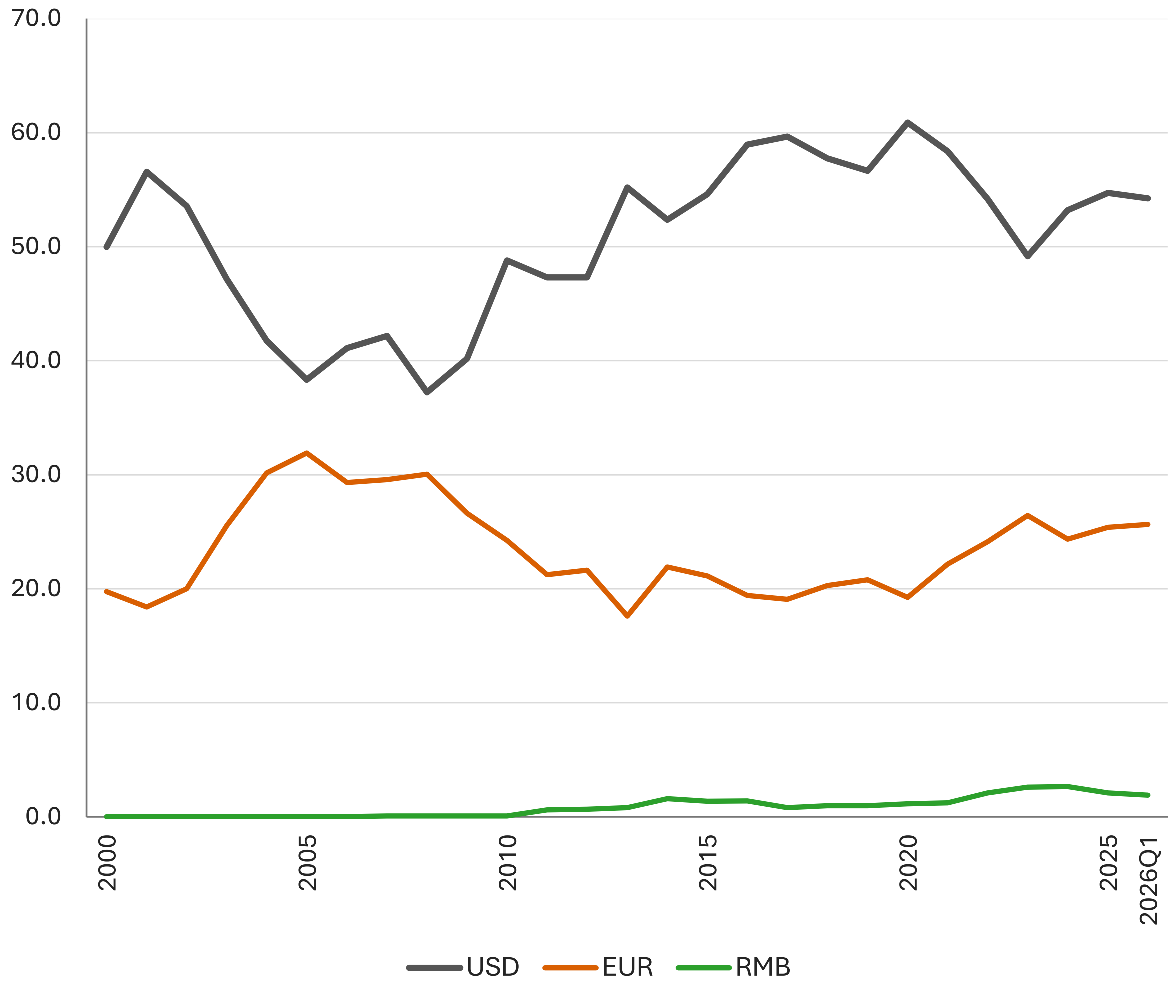
International Debt Securities

(currency shares in percent)

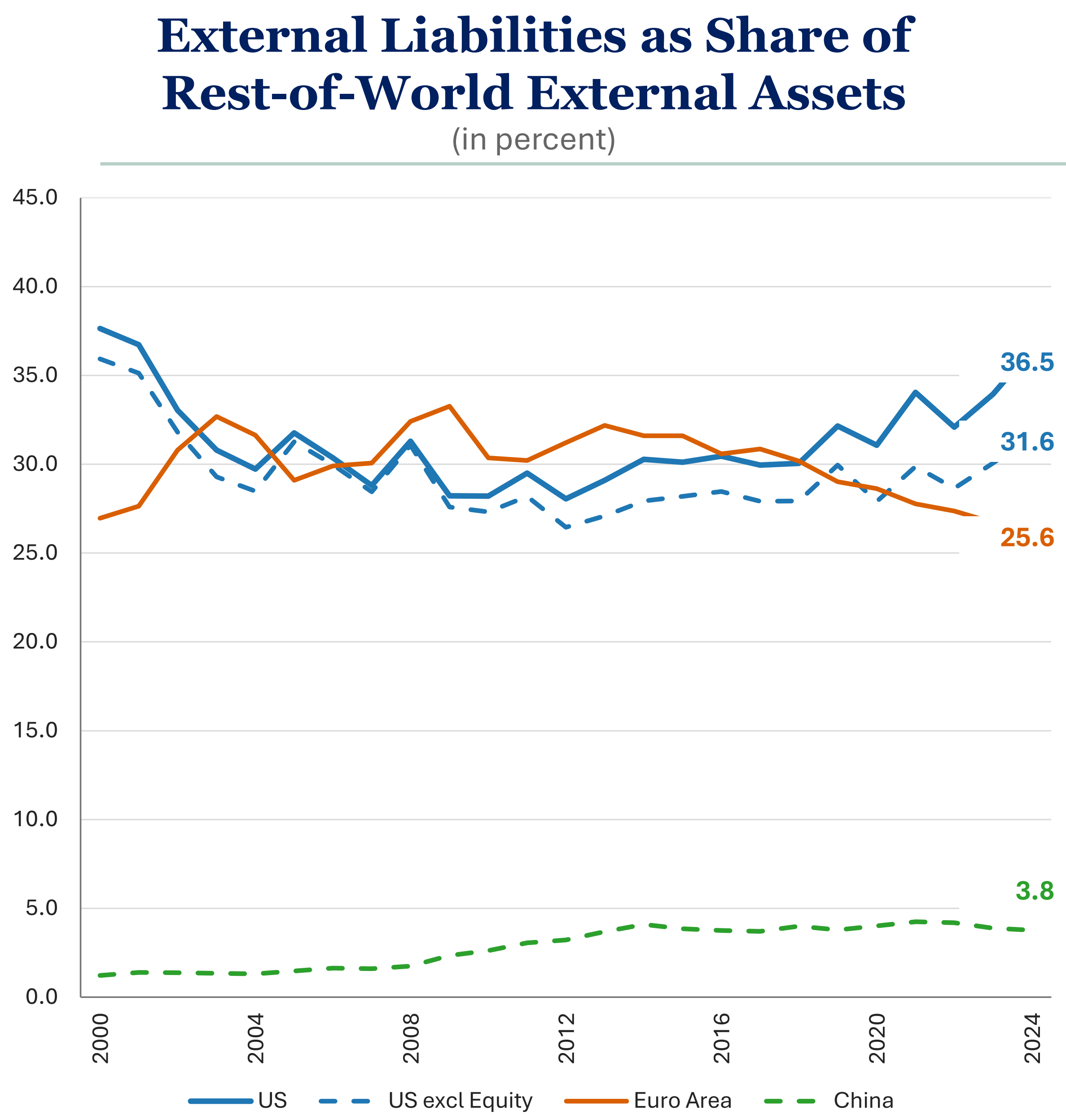
Outstanding Stocks

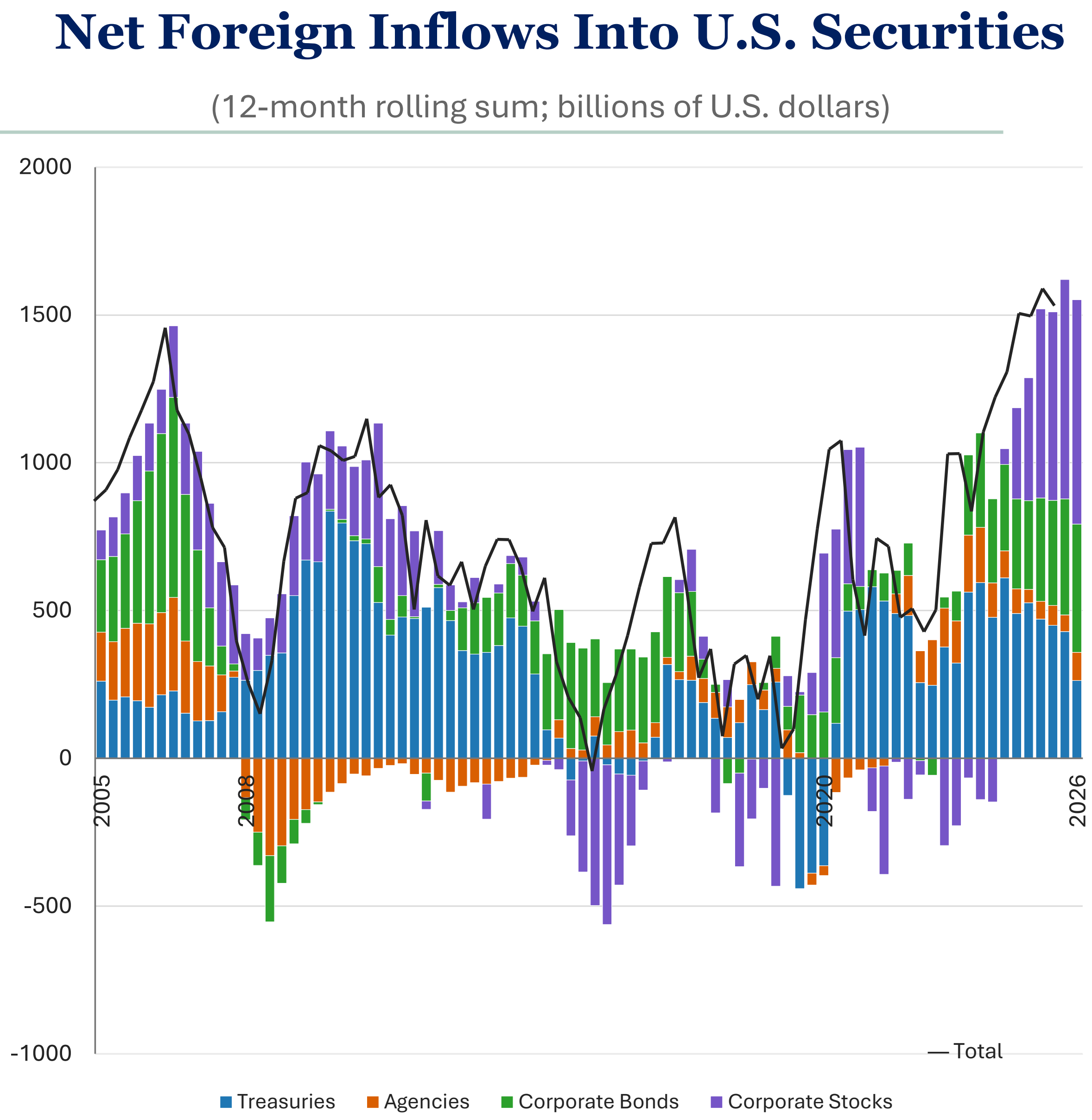


Gross Issuances



Source: Pradhan et al. (2026)


External Liabilities as Share of Rest-of-World External Assets
(in percent)
45.0
40.0
35.0
30.0
25.0
20.0
15.0
10.0
5.0
0.0
2000
2004
2008
2012
2016
2020
2024
36.5
31.6
25.6
3.8
US
US excl Equity
Euro Area
China

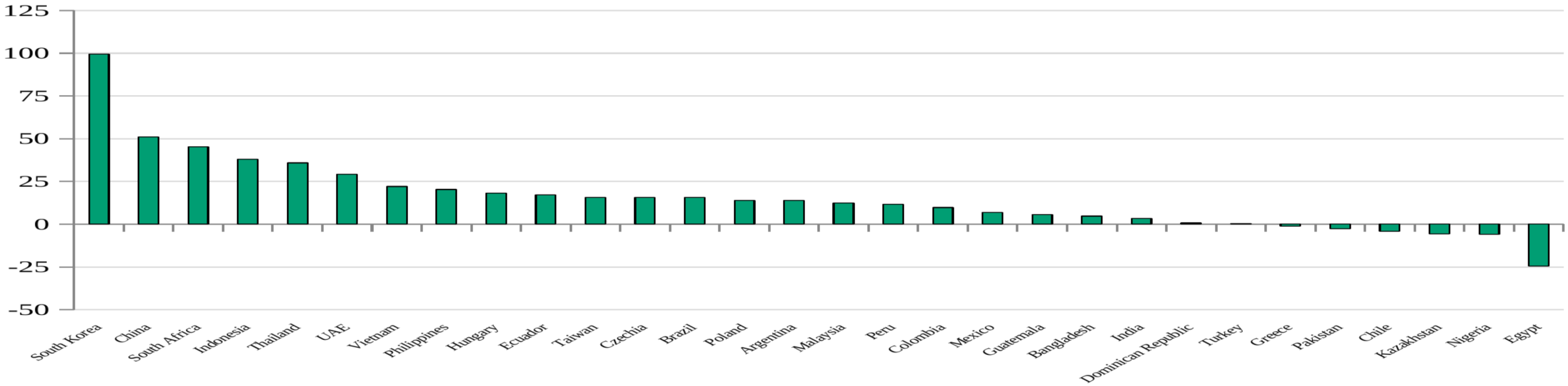

Net Foreign Inflows Into U.S. Securities
(12-month rolling sum; billions of U.S. dollars)
2000
1500
1000
500
0
-500
-1000
2005
2008
2020
2026
Total
Treasuries
Agencies
Corporate Bonds
Corporate Stocks

Safe Asset Demand From EMDEs and Small Open Economies

- EMDEs becoming more financially open, greater need for “insurance”
- Reserve accumulation by central banks has leveled off
- Ratio of (External Assets minus Reserves) to External Liabilities rising
- Private holdings of Treasuries up, official holdings flat
- EMDE private holdings of U.S. securities rising

Self-Insurance Through Private Sector Accumulation of Foreign Assets

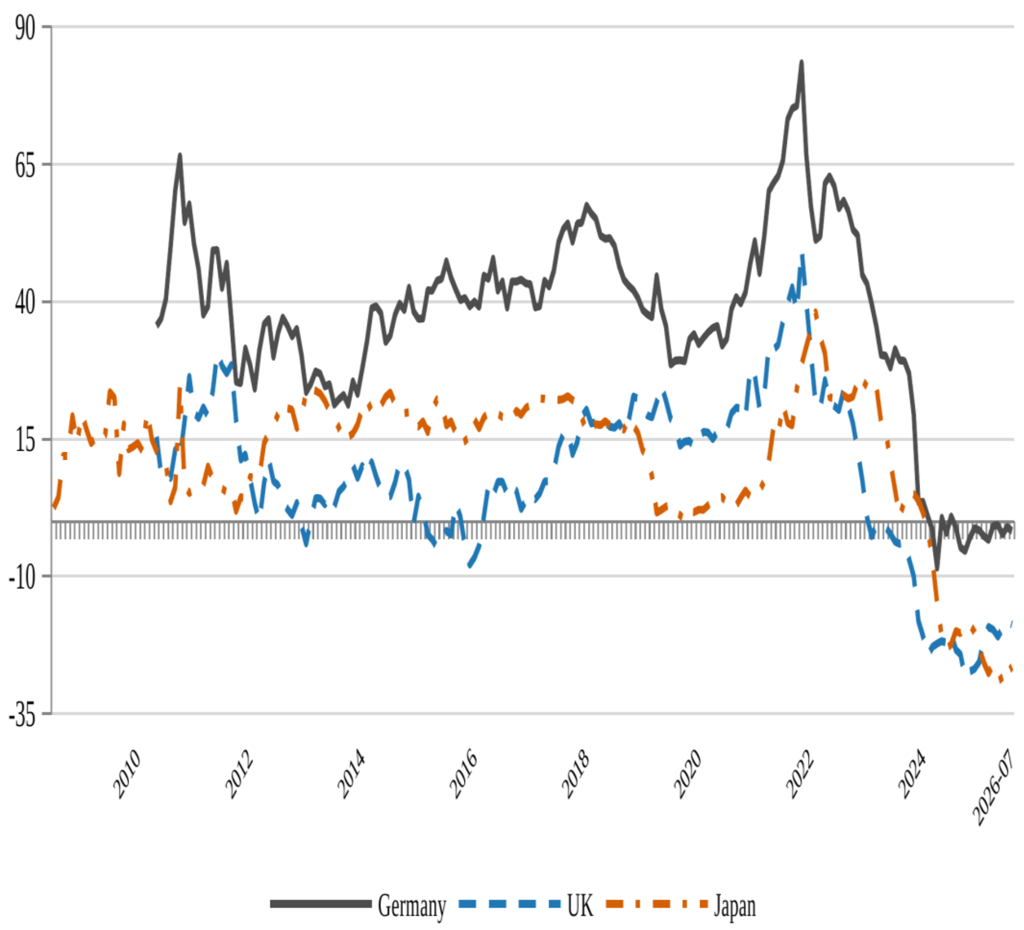
(Change from 2010 to 2024; Percentage Points)



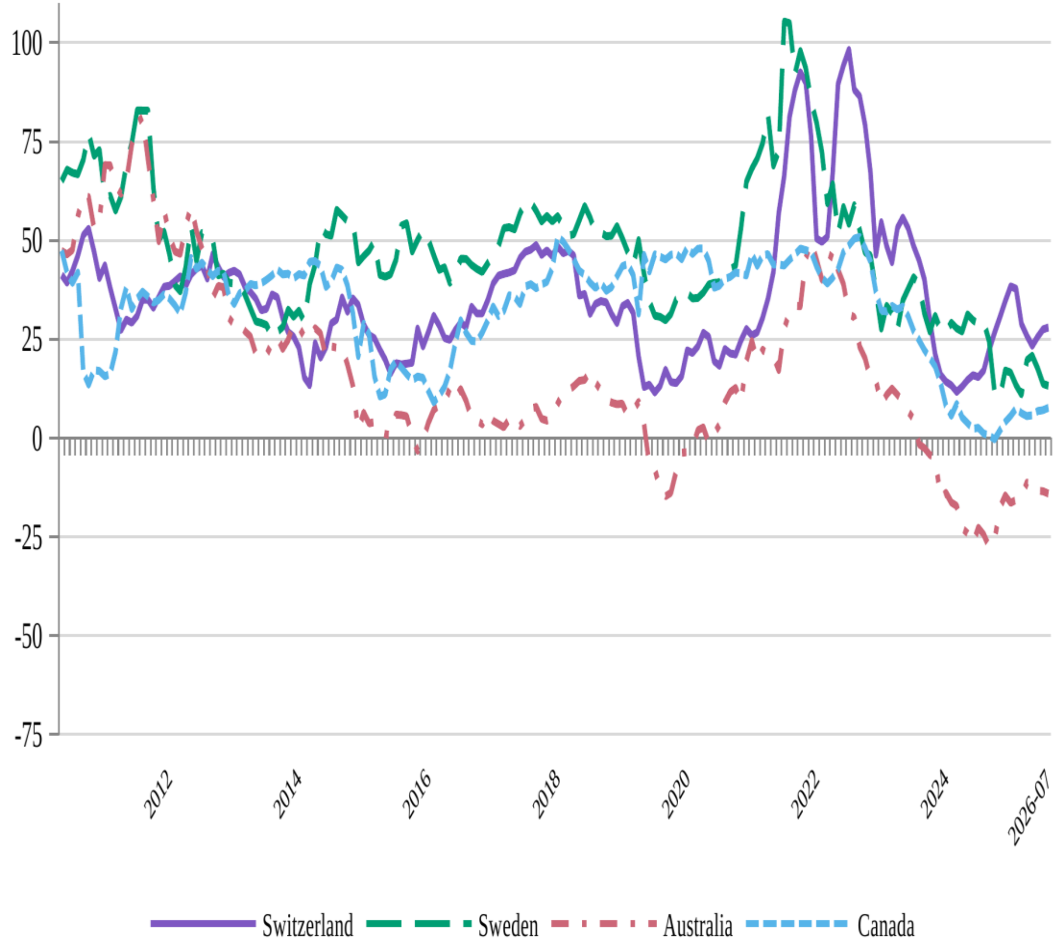
Diverging Market Perceptions of Safety

(estimated convenience yields on 10-year government bonds, in basis points)

Germany, UK, Japan

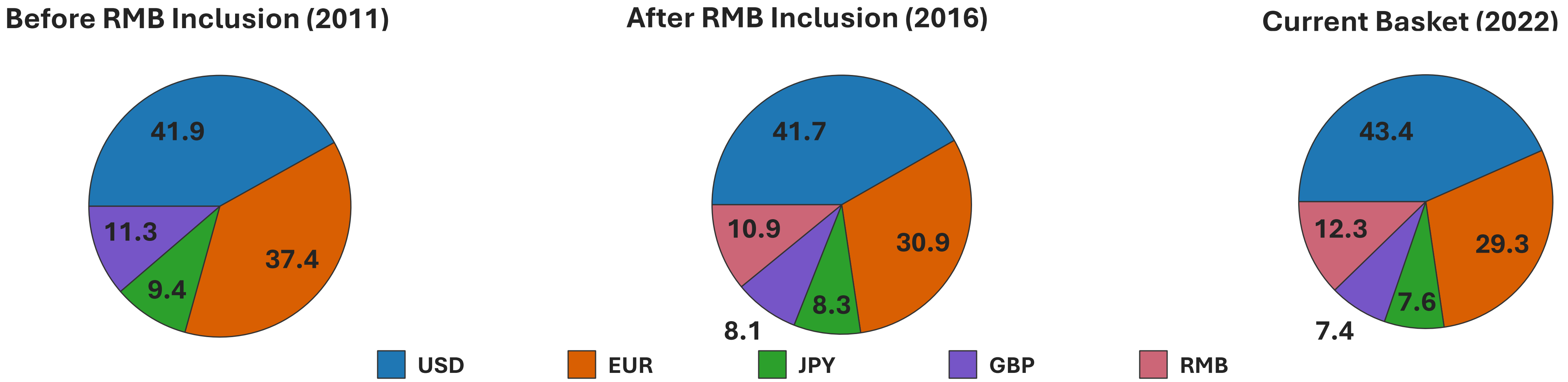


Switzerland, Sweden, Australia, Canada



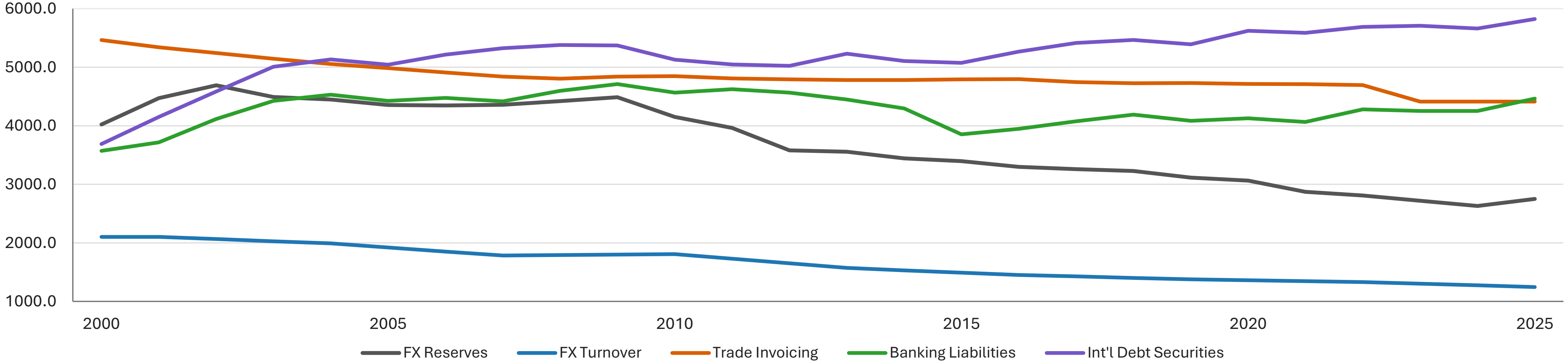
SDR Composition

(currency shares, in percent)



Multipolarity or Fragmentation?

(Herfindahl Index; Dollar Excluded)



Will Financial Innovation Reshape the IMS?

- Financial innovation
  - Domestic fast payment systems
  - Cross-border payments: Fintech platforms, W-CBDC bridges
  - Private currencies: decentralized cryptocurrencies, stablecoins
- Consequences
  - Efficiency, speed, cost of cross-border transactions (payments, asset trade)
  - Fewer frictions, more conduits => more capital flow, FX volatility
  - *Outcome A*: More level playing field, lesser need for vehicle currencies
  - *Outcome B*: Greater concentration at top, fragmentation in second tier, loss of sovereignty in lower tiers
  - *Bifurcation of functions of money*: medium of exchange / store of value

Modeling Currency Competition / Dominance

- Existing theories

  A. Invoicing–reserve complementarity (Gopinath–Stein 2021; Chahrour–Valchev 2021)

  B. Unit-of-account coordination (Doepke–Schneider 2017)

  C. Liquidity-based debt denomination (Coppola–Krishnamurthy–Xu 2026)

- We build on C

  Firms search for settlement in both local, global markets

  (local search not frictionless outside option)

  Introduce stablecoins as innovation in settlement technology

  (proxy for innovations that reduce frictions)

  Settlement technology as margin of competition

  Interaction of market depth, liquidity, and settlement technology

The Friction: Debts Must Be Settled in Denomination Currency

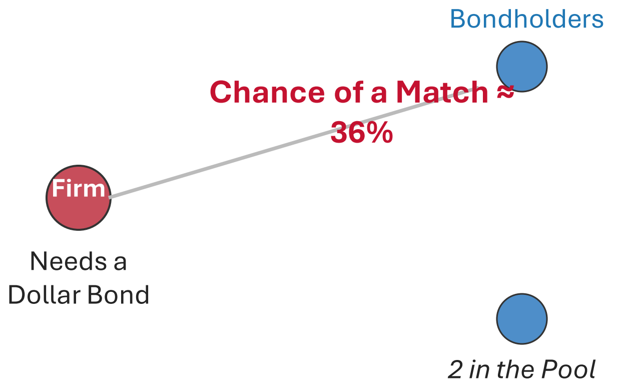
t_0	t_1	t_2
Firm issues debt in chosen currency (local or dollars) to fund project	Settlement need arises (prob. ϕ): firm must obtain safe claim in currency it owes	Debt due, extinguished with the claim in hand

Cash flows and the obligation are not synchronized: revenue may arrive before or after due date

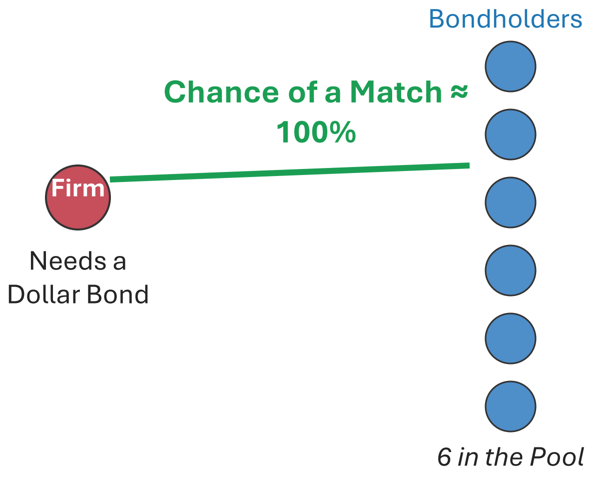
If firm owes dollars but cannot obtain dollar claim at t_1, it settles in local-currency cash — a costly outcome

Settling a dollar debt requires search: Firm must find a bondholder

Thin Market — Few Bonds to Find



Deep Market — Easy to Find One



Stablecoin

New settlement technology: Claim redeemable at par in settlement asset

Search-free, certain, access cost $\delta \geq 0$

Its demand is rerouted to the bond market through the issuer's reserves — not removed.

Key Properties

- Issuance: how easily firms settle
- Settlement demand: how easily bonds resell

⇒ *Can price currency's convenience separately from depth*

- Settlement by search has two costs
  - Search is costly
  - Settling without a safe claim is costly

- Firm-side probability increasing in depth
  - Depth includes endogenous issuance
  - Settlement lqdy improves as more firms adopt currency

⇒ *Thick-market externality*

- Bondholder-side probability rises with settlement demand
- Settlement certainty drives concentration in currency that backs stablecoin

EME Firms' Currency Choice

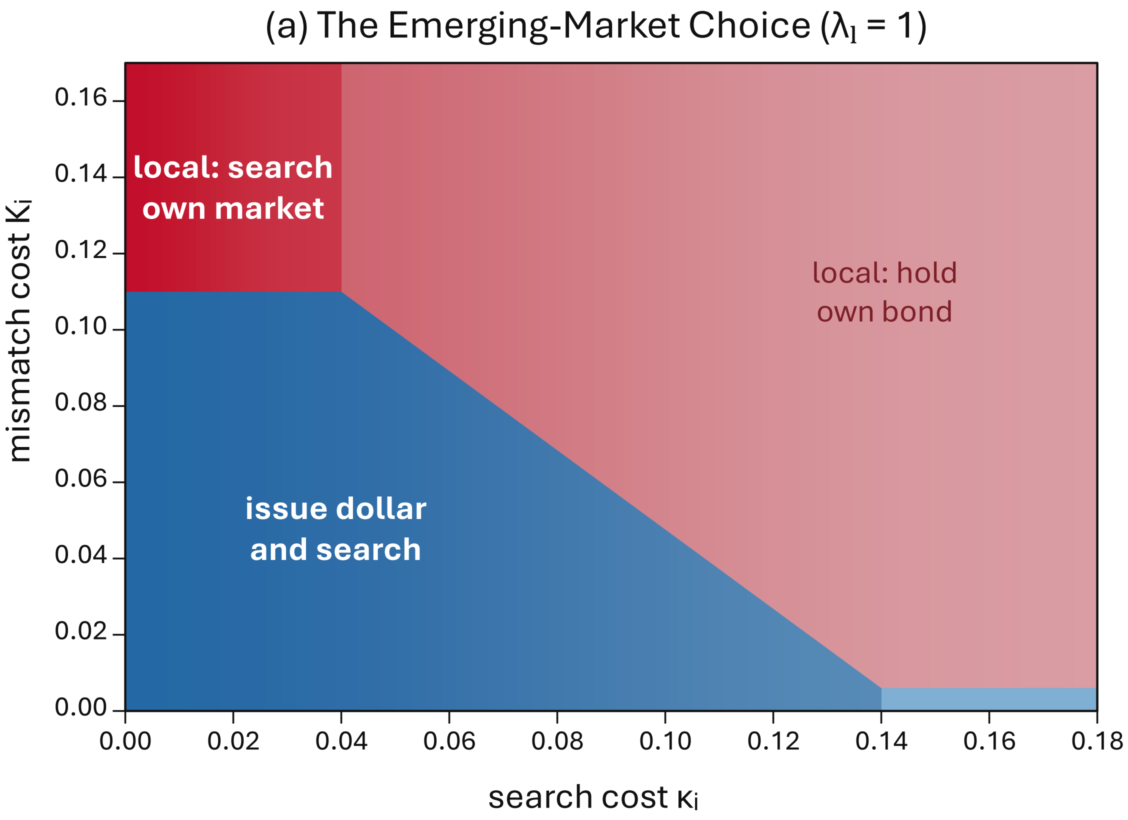
- **Two-Step Decision**
  - Issuance currency: Local or global
  - Settlement route: Search, stablecoin, or cash
- Issuance premium, settlement probability common across firms
- Favors deeper, more liquid markets

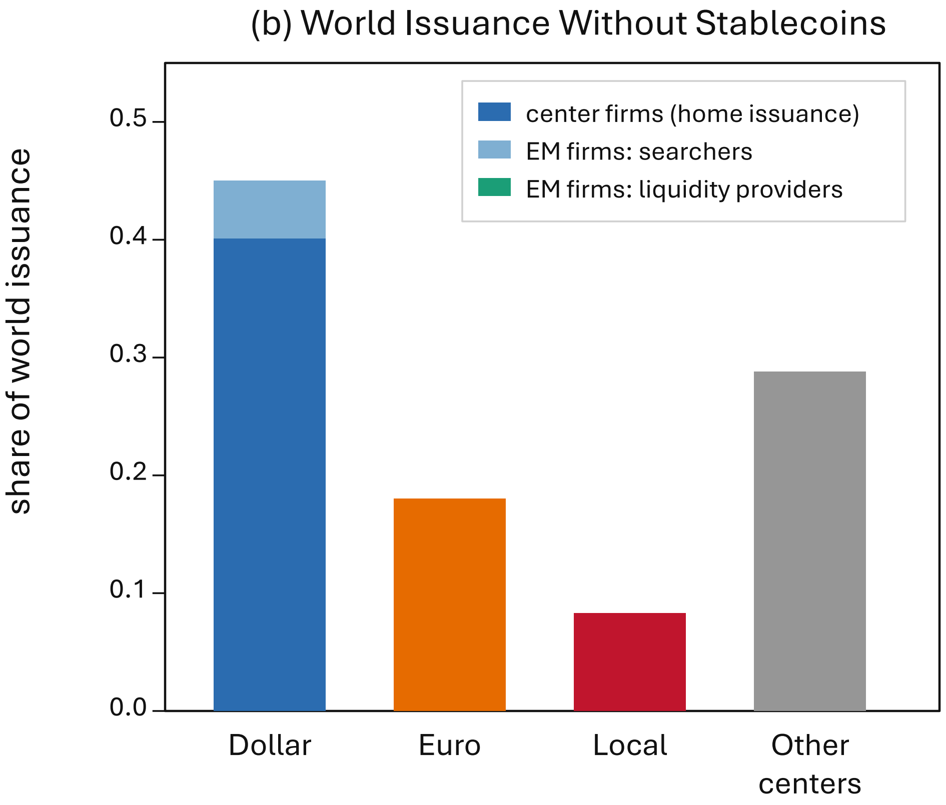
⇒ *Concentration in one market*

⇒ *Self-reinforcing: Issuance deepens the market and eases settlement*

Result 1 — Innovation in EMEs Promotes Local Currency

Each EME firm chooses: Issuance currency — local (avoids mismatch cost) or global
Settlement route — search or hold cash

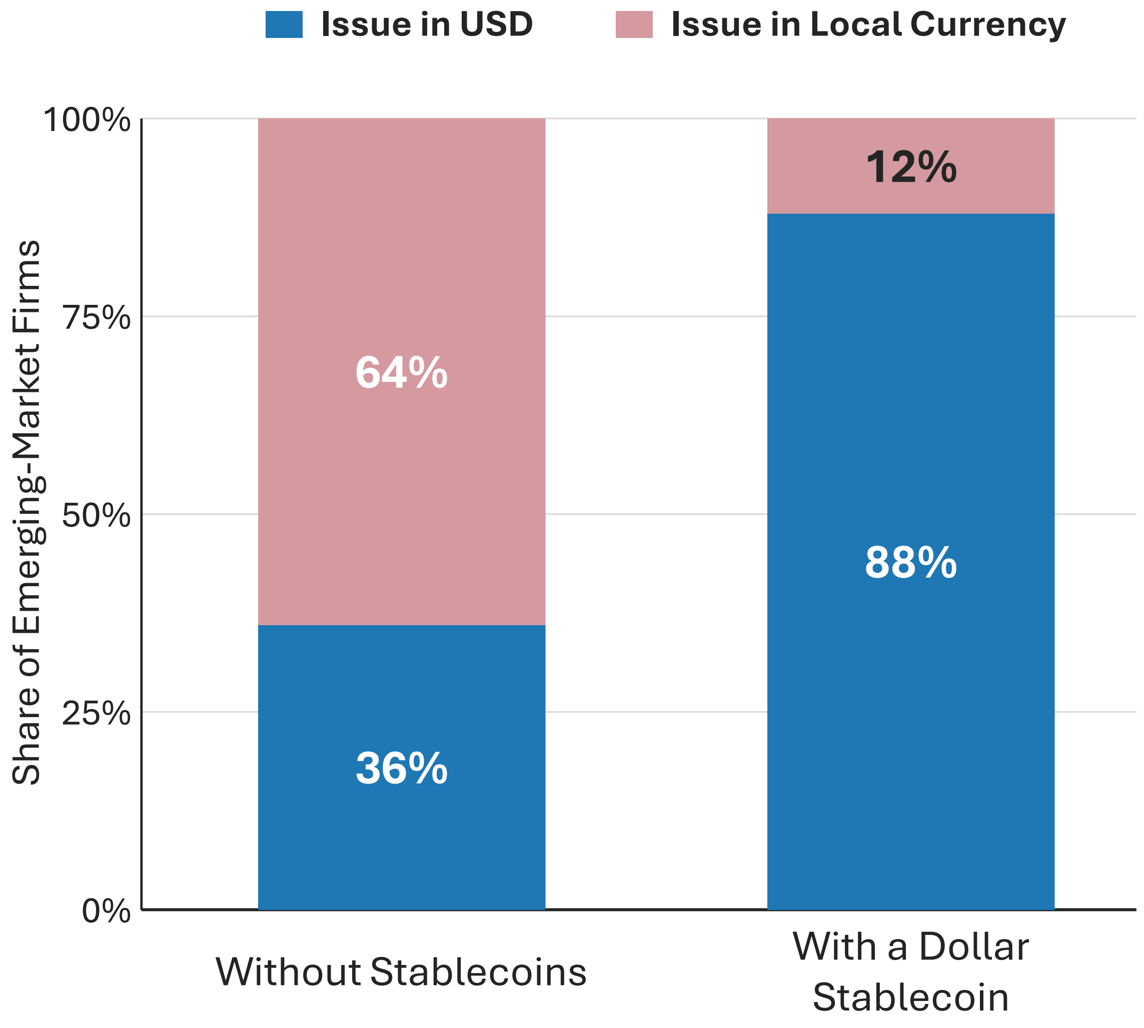




Raising local market's matching efficiency (λ_l) or deepening its float (L_l) increases local-currency settlement odds and its liquidity premium, expanding the local regions in (a).

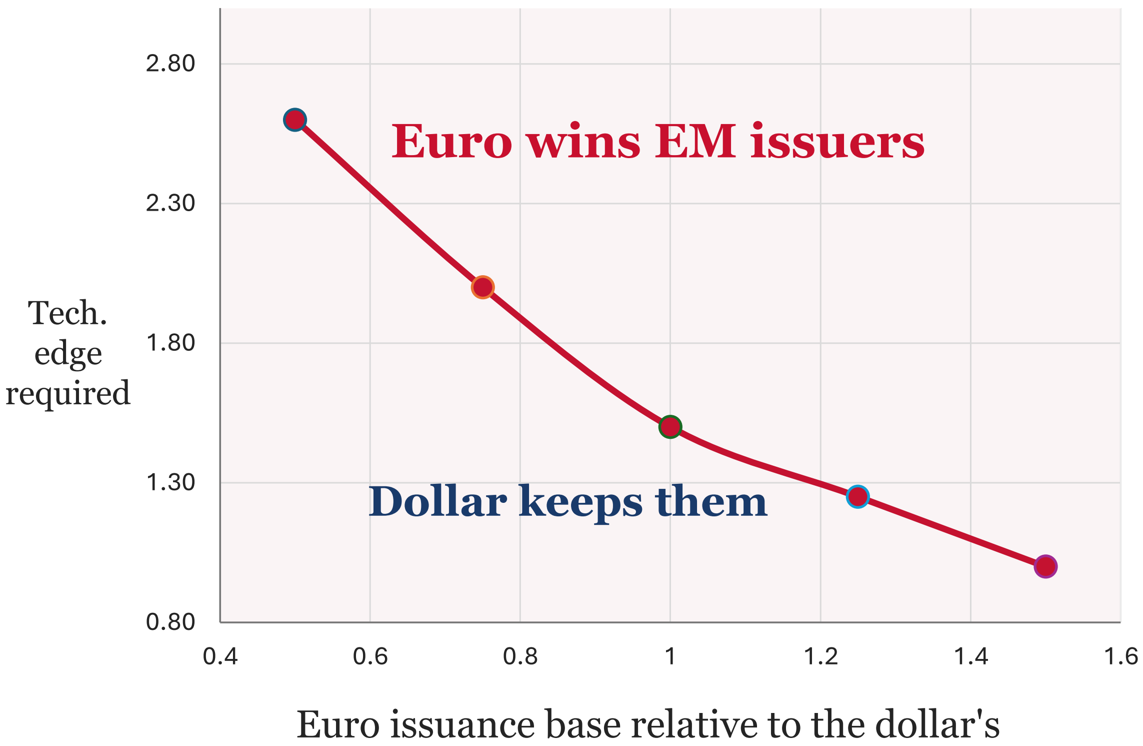
Currency Competition: Market Depth and Technology

Result 2 — Dollar Stablecoin Deepens Dollar Use



Result 3 — Global Currency Competition

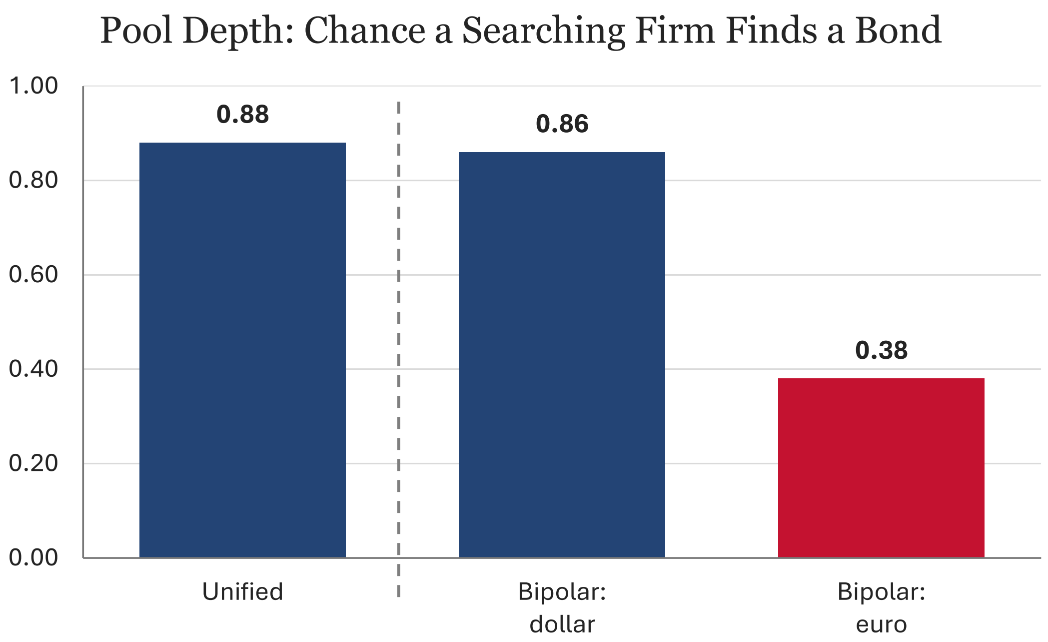
Benchmark calibration — illustrative magnitudes



At a 1.25× issuance base, no technology edge is needed.

Fragility: Fragmentation Thins Markets, Concentrates Crisis Losses

Fragility I — Fragmentation Thins Every Market



Both fragmented-market pools are thinner

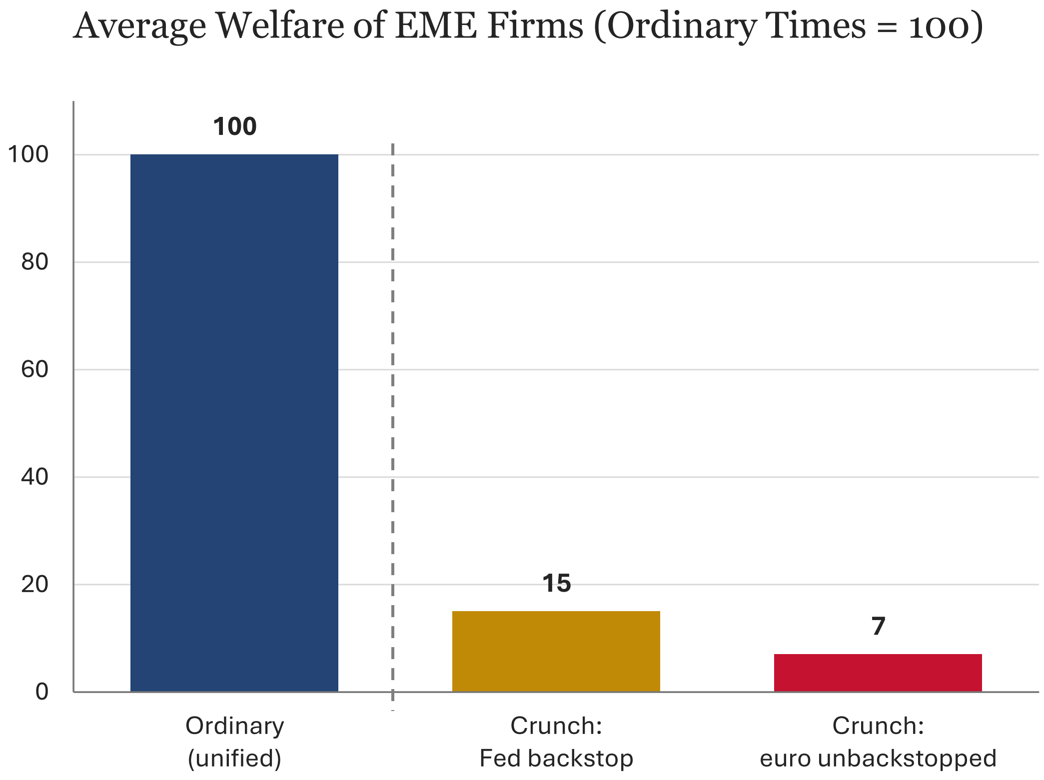
Average Welfare of EME Firms

100 → 49

unified bipolar

Fragility II — Liquidity Crunch Worse in Fragmented Market

(benchmark calibration — illustrative magnitudes)



The crunch is costly even fully backstopped
Fragmentation roughly halves what remains

Structure, Fragility of IMS Not Preordained

- Three worlds to contemplate:
  - 1: *Greater concentration*, with fragmented second tier
  - 2: *Multipolarity*, with strong institutions & regulation, deep & liquid markets
  - 3: *Fragmentation*, built on wobbly foundations => competition breeds fragility

- Policy implications
  - Improve settlement technology, build market depth
  - Strengthen macro, regulatory, institutional policies and frameworks
  - Structure, credibility of liquidity backstops crucial; cross-border design matters

Search, Matching, and Stablecoins (Details)

Search and Matching

Obtaining a bond at t1 requires search: firm must find bondholder willing to sell

$\alpha_{I,c} = \lambda_c \phi (S_c + M_c^B)$ mass of firms searching currency c's money market at t1

M_c^B mass of bondholders

$$\alpha_{F,c} = \lambda_c (L_c + M_c)$$

government bond float + private issuance

Introduction of Stablecoins

$$\alpha_{I,c} = \lambda_c \phi (S_c + M_c^B)$$

bondholder-side meeting probability

M_c^B mass of currency-c issuers who settle by stablecoin

Stablecoin demand still reaches bondholders through the issuer's reserve purchases: the demand is rerouted, not removed.

Matching Technology

$$\alpha_{I,c} = \lambda_c \phi S_c$$

S_c	$m_{F,c} = \phi S_c$	$\alpha_{I,c} = \lambda_c \phi (S_c + M_c^B)$
matching efficiency	*increasing returns*	*Poisson benchmark*

Two Sides of the Market

With θ = 1, each side's meeting probability depends on the mass on the other side

Searching firm finds bond

rate set by depth of pool it searches

$$\alpha_{F,c} = \lambda_c (L_c + M_c)$$

pool depth

Bondholder meets buyer

mass of firms searching the market

$$\alpha_{I,c} = \lambda_c \phi S_c$$

settlement demand

S_c mass of currency *c* issuers who settle by search

$$m_{F,c} = \phi S_c$$